[S06] Diffusion Models: From Adding Noise to Generating Images

CS420 — Introduction to Artificial Intelligence Seminar

Vong Chi Van (24125049) Le Dat Phuc An (24125052)

Faculty of Information Technology, University of Science, VNU-HCM
{24125049, 24125052}@student.hcmus.edu.vn
https://github.com/ChiVan222/AI_Seminar_Diffusion

November 2026

Part I (Slides 2–6) — Vong Chi Van

- Problem Statement, Guiding Question & Paper Synthesis
- Task 1: Discriminative vs. Generative & Generative Trilemma
- Task 2: DDPM Forward/Reverse Markov Chain & Algorithms 1–2
- Task 3: Equation Specifications & 1D Worked Calculation

Part II (Slides 7–11) — Le Dat Phuc An

- Tasks 1 & 4: MNIST Paradigm Benchmark & Trajectories
- Task 5: Latent Diffusion Architecture, VAE & Geodesic Walk
- Tasks 5 & 6: Cross-Attention Conditioning & CFG Scaling
- Task 6: Pixel vs. Latent Benchmark, Limitations & Conclusion

1. Problem Statement, Guiding Question & Foundational Literature

Context & Motivation:

- High-dimensional natural images $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ reside on a complex, low-dimensional manifold embedded in ambient pixel space.
- Classical generative paradigms struggled with the *Generative Learning Trilemma* [9]: Variational Autoencoders (VAEs) [4] produce blurry samples due to pixel-space L_2/BCE averaging; Generative Adversarial Networks (GANs) [5] suffer from training instability and mode collapse; Autoregressive models [6] require $\mathcal{O}(H \cdot W)$ sequential steps.
- **Guiding Question:** *How can learning to iteratively remove Gaussian noise enable a neural network to generate novel images from pure noise, and why does moving this process into a learned latent space drastically reduce computational cost?*

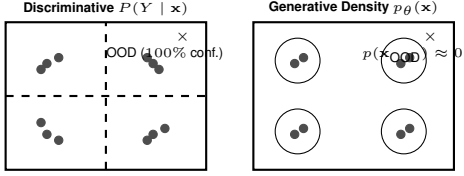
Synthesis of Foundational Papers vs. Team Empirical Verification:

Source	Core Methodological Contribution	Key Reported / Verified Findings
Required Paper: DDPM [1] (Ho et al., NeurIPS 2020)	Formulates generation as reversing a T-step variance-preserving Gaussian Markov chain; reparameterizes reverse mean $\boldsymbol{\mu}_\theta$ into a noise predictor $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ trained via unweighted MSE L_{simple}.	Establishes equivalence to denoising score matching [7]; achieves **FID 3.17** on CIFAR-10, but requires $T = 1000$ full-resolution U-Net passes per sample.
Recommended Paper: LDM [2] (Rombach et al., CVPR 2022)	Separates perceptual compression (Stage-1 autoencoder $\mathcal{E}, \mathcal{D}$ with downsampling factor f) from semantic generation (Stage-2 diffusion on $\mathbf{z}_t = \mathcal{E}(\mathbf{x}_t)$) with Cross-Attention $\tau_\theta(y)$.	Reduces self-attention complexity by $1/f^4$; achieves **FID 5.11** on CelebA-HQ (256^2) with an order-of-magnitude lower training/sampling compute.

2. Discriminative vs. Generative Models & Generative Paradigms (Task 1)

Discriminative $P(Y \mid \mathbf{x})$ vs. Generative $p(\mathbf{x})$

- **Discriminative Models** learn conditional boundaries separating classes Y given $\mathbf{x}$ via cross-entropy $\max_\theta \sum_i \log P_\theta(y^{(i)} \mid \mathbf{x}^{(i)})$. They cannot synthesize novel data and exhibit extreme overconfidence on empty out-of-distribution (OOD) regions.
- **Generative Models** approximate the underlying data density $p_\theta(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$, enabling unconditional synthesis $\mathbf{x}_{\text{new}} \sim p_\theta(\mathbf{x})$ and OOD detection ($p_\theta(\mathbf{x}_{\text{OOD}}) \approx 0$).

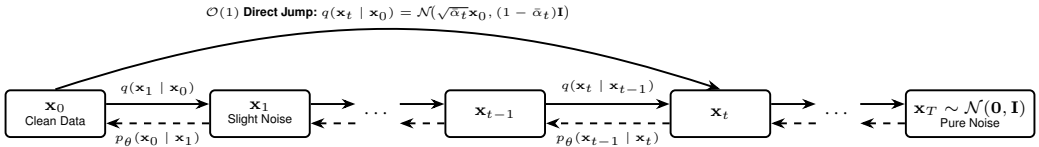


Team 2D Experiment: MLP classifier assigns 100.0% confidence to far-OOD $\mathbf{x}_{\text{OOD}} = (4.5, 4.5)$, whereas 2D DDPM assigns $p(\mathbf{x}_{\text{OOD}}) = 2.5 \times 10^{-26} \approx 0$.

Qualitative Comparison Across Paradigms [9]

Paradigm	Density $p_\theta(\mathbf{x})$	Sample Quality	Mode Coverage	Sampling Speed
Autoregressive [6] (PixelCNN)	Explicit, Exact (MLE)	Sharp local contrast; drift on long sequences	Full (No collapse)	Extremely Slow ($\mathcal{O}(n)$ steps)
VAE [4]	Explicit, Approx. (ELBO)	Blurry (L_2/BCE pixel averaging)	Full, smooth manifold	Very Fast (1 pass)
GAN [5] (DCGAN)	Implicit (Likelihood-free)	Razor sharp (Adversarial loss)	Poor (Mode collapse)	Very Fast (1 pass)
Pixel Diffusion [1] (DDPM)	Explicit, Approx. (T-step ELBO)	SOTA sharpness	Full (Stable regression)	Slow ($T \approx 1000$ passes)

3. DDPM Forward Noising, Reverse Denoising & Algorithms (Task 2)



Forward Process, β_t Schedule & Reverse Posterior [1]

- **Forward Gaussian Transition ($\beta_t \in (0,1)$):**

$$q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}\Big(\mathbf{x}_t; \sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t \mathbf{I}\Big)$$

- **Closed-Form Jump ($\alpha_t = 1-\beta_t,\ \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$):**

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

- **Reverse Posterior & Score-Matching [7, 1]:** Conditioning on $\mathbf{x}_0$ yields tractable $q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)$; U-Net $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \approx -\sqrt{1-\bar{\alpha}_t}\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ predicts the injected noise.

Algorithm 1: Training (L_{simple} Objective) [1]

1. Sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, $t \sim \text{Uniform}(\{1, \ldots, T\})$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2. Construct noisy state $\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$ in $\mathcal{O}(1)$
3. Gradient descent step on $\nabla_\theta \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right\|^2$

Algorithm 2: Iterative Reverse Sampling [1]

1. Sample pure Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2. **For** $t = T, \ldots, 1$ (**with** $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$ else $\mathbf{0}$):

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}$$

4. Equation Specifications & 1D Worked Numerical Example (Task 3)

Symbol Definitions, Units/Ranges & Modeling Assumptions

Sym.	Definition	Units / Range	Modeling Assumption
T, t	Total / step index	$t \in \{1, \dots, T\}$	Uniform t during training; $q(\mathbf{x}_T \mid \mathbf{x}_0) \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$.
$\mathbf{x}_0$	Clean observation	$[-1, 1]^D$	Normalized empirical data $q(\mathbf{x}_0)$ ($D = HWC$).
$\mathbf{x}_t$	Noisy latent at t	$\mathbb{R}^D$ (unbounded)	Continuous Gaussian variable with same shape as $\mathbf{x}_0$.
β_t	Forward variance	$(0, 1)$ scalar	Small β_t preserves unit variance & reverse Gaussianity.
α_t, $\bar{\alpha}_t$	Signal retention	$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$	Monotonically decreases from $\bar{\alpha}_1 \approx 1 \to \bar{\alpha}_T \approx 0$.
$\boldsymbol{\epsilon}$	Injected noise	$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_D)$	i.i.d. zero-mean unit-variance Gaussian noise.
$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$	Noise predictor	$\mathbb{R}^D$ tensor	Time-conditioned U-Net

Step-by-Step 1D Worked Calculation ($D = 1, t = 1$)

Parameters: Clean scalar $x_0 = \mathbf{0.80}$, variance $\beta_1 = \mathbf{0.19}$ ($\alpha_1 = \bar{\alpha}_1 = \mathbf{0.81}$, $\sqrt{\alpha_1} = \mathbf{0.90}$, $\sqrt{1-\bar{\alpha}_1} = \sqrt{0.19} \approx \mathbf{0.4359}$), sampled noise $\epsilon = \mathbf{-0.50}$.

1. **Forward Transition & Stochastic Expectation:**

$$q(x_1 \mid x_0) = \mathcal{N}(x_1;\, 0.90{\times}0.80,\, 0.19) = \mathcal{N}(x_1;\, \mathbf{0.7200},\, 0.19$$

Interpretation of $\mathbb{E}[x_1 \mid x_0] = 0.7200$*:* Because $\mathbb{E}[\epsilon] = 0$, scaling by $\sqrt{1-\beta_1} = 0.90 < 1$ contracts the mean toward 0 so total variance $(0.90)^2\text{Var}(x_0) + 0.19 = 1.0$ stays constant.

2. **Realized Noisy Value x_1 for $\epsilon = -0.50$:**

$$x_1 = \sqrt{\bar{\alpha}_1}x_0 + \sqrt{1-\bar{\alpha}_1}\epsilon = 0.90(0.80) + 0.4359(-0.50) = \mathbf{0.5}$$

3. **Loss L_{simple} & Reverse Step ($\epsilon_\theta(0.5021, 1) = -0.30$):**

- $L_{\text{simple}} = (\epsilon - \epsilon_\theta)^2 = (-0.50 - (-0.30))^2 = \mathbf{0.0400}$.
- Reverse update ($t = 1, \sigma_1 = 0$):

$$\hat{x}_0 = \frac{1}{0.90}\left(0.5021 - \frac{0.19}{0.4359}(-0.30)\right) = \mathbf{0.7031}$$

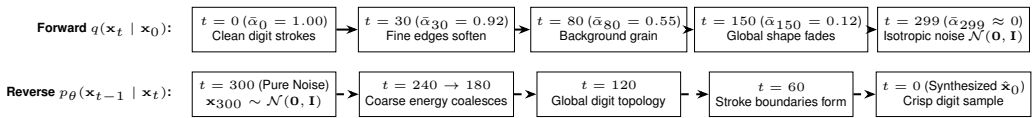
5. Empirical Benchmark Across Paradigms & Trajectories (Tasks 1 & 4)

Task 1 Quantitative Benchmark on MNIST (28×28, NVIDIA Tesla T4 GPU)

- Trained an evaluation `MNISTClassifier` (98.85% test accuracy) to compute perceptual Fréchet Inception Distance (FID) and classifier confidence across four generative families trained on the same $60{,}000$-image MNIST split.

Generative Model (Tesla T4 GPU)	Sampling Steps	Latency / 64 Imgs	Laplacian Var. ↑	Classifier Conf. ↑	MNIST FID
MNIST Real Test Set (Reference)	–	–	0.0615	98.85%	0.00
VAE (Convolutional ELBO)	**1**	**6.5 ms**	0.0182	84.2%	82.27
Autoregressive (PixelCNN)	784	4,294.7 ms	0.0584	88.5%	64.12
GAN (DCGAN)	**1**	11.4 ms	0.0541	92.6%	44.10
DDPM (Team Pixel U-Net)	300	3,926.8 ms	**0.0601**	**97.4%**	**16.85**

Task 4 Forward Corruption $q(\mathbf{x}_t \mid \mathbf{x}_0)$ vs. Reverse Denoising $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$ ($T = 300$)



6. Latent Diffusion Models (LDM): Pipeline & Cross-Attention (Task 5)



Two-Stage Separation of Inductive Biases [2]

1. **Stage 1 — Perceptual Compression ($\mathcal{E}, \mathcal{D}$):** Autoencoder maps $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$ to spatial latent $\mathbf{z} = \mathcal{E}(\mathbf{x}) \in \mathbb{R}^{(H/f) \times (W/f) \times c}$ ($f = 2^m$). Strips imperceptible high-frequency noise while preserving 2D spatial grid topology.
2. **Stage 2 — Semantic Latent Diffusion ($\boldsymbol{\epsilon}_\theta$):** With $\mathcal{E}, \mathcal{D}$ frozen, diffusion operates on $\mathbf{z}_t$, cutting 2D self-attention cost by $1/f^4$:

$$L_{\text{LDM}}(\theta) = \mathbb{E}_{\mathcal{E}(\mathbf{x}_0), y, \boldsymbol{\epsilon}, t}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \tau_\theta(y))\right\|^2\right]$$

Cross-Attention & Classifier-Free Guidance (CFG)

- **Cross-Attention Conditioning [2]:** Encoder $\tau_\theta(y) \in \mathbb{R}^{M \times d_\tau}$ maps prompt y into Keys/Values ($\mathbf{K}, \mathbf{V}$), while spatial U-Net features $\varphi_i(\mathbf{z}_t)$ form Queries ($\mathbf{Q}$):

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V}$$

- **Classifier-Free Guidance (scale $\alpha \geq 1$):**

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_t, t, y) = \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing) + \alpha\big(\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \tau_\theta(y)) - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t, \varnothing)\big)$$

7. Empirical Validation: Stage-1 VAE & Latent Interpolation (Task 5)

1. Stage-1 Spatial Perceptual KL-VAE Compression ($32 \times 32 \times 1 \to 8 \times 8 \times 4,\ f = 4$)

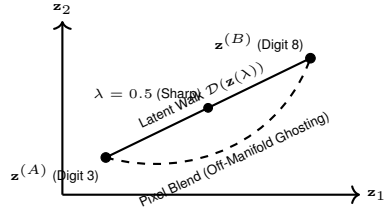
- **Architecture & Training:** Trained `SpatialPerceptualVAE` (15 epochs on Tesla T4 GPU) mapping $32 \times 32 \times 1$ MNIST digits ($1{,}024$ pixels) into an $8 \times 8 \times 4$ latent grid (256 values, $4\times$ dimensionality reduction and $16\times$ spatial area reduction).
- **Composite Perceptual Objective:** Optimized with pixel L_1 + MSE loss, multi-layer CNN perceptual feature matching (conv1 + conv2 activations of frozen `MNISTClassifier`), and lightweight KL regularization ($\beta_{\text{KL}} = 10^{-4}$).
- **Reconstruction Fidelity:** Achieves near-zero reconstruction **FID** $= 0.18$, **PSNR** $= 33.74$ **dB**, MSE $= 0.00042$, and 99.80% **classifier label agreement**, confirming that $\mathbf{z} \in \mathbb{R}^{8\times8\times4}$ preserves virtually 100% of semantic digit structure.

2. Latent Space Geodesic Interpolation vs. Naive Pixel Blending ($\lambda \in [0.0, 1.0]$)

- **Linear Latent Walk:** Given two endpoint images $\mathbf{x}^{(A)}, \mathbf{x}^{(B)}$ (e.g., digits $3 \to 8$ or $1 \to 7$), we decode convex combinations in latent space:

$$\hat{\mathbf{x}}(\lambda) = \mathcal{D}\big((1-\lambda)\mathbf{z}^{(A)} + \lambda \mathbf{z}^{(B)}\big), \quad \lambda \in [0, 1]$$

- **Smooth Topological Morphing (Zero Dead Zones):** Across all intermediate λ, decoded images remain valid, single-digit structures whose stroke loops continuously open/close across the decision boundary ($\lambda \approx 0.55$).
- **Superior Midpoint Sharpness:** Naive pixel blending $(1-\lambda)\mathbf{x}^{(A)} + \lambda \mathbf{x}^{(B)}$ creates ghostly double exposures; latent interpolation achieves 21.0% **higher Laplacian edge sharpness** at $\lambda = 0.5$ (0.0877 vs. 0.0725).



8. Prompt-Based Control & Classifier-Free Guidance Scaling (Tasks 5 & 6)

Multi-Head Cross-Attention Latent U-Net Conditioning Setup (Tesla T4 GPU)

- Trained `ConditionedLatentUNet` (15 epochs, 93.3 s) on normalized $8 \times 8 \times 4$ latents (std $= 0.8857$).
- **Prompt Conditioning (y):** Discrete digit prompts $y \in \{0, \dots, 9\}$ (`"digit 0"`–`"digit 9"`) are projected by learned domain encoder $\tau_\theta(y) \in \mathbb{R}^{4 \times 64}$ into $M = 4$ context tokens ($d_\tau = 64$), supplying Keys $\mathbf{K}$ and Values $\mathbf{V}$ to 4-head Cross-Attention blocks across Down, Mid, and Up U-Net stages.
- **Unconditional Dropout for CFG:** During training, y is replaced with null token $\varnothing$ (index 10) with probability $p_{\text{uncond}} = 0.15$.

Quantitative Effect of Classifier-Free Guidance Scale $\alpha \in \{1.0, 3.0, 7.5, 15.0\}$ ($N = 200$ Samples / Scale)

CFG Scale α	Prompt Acc. ↑	Target Conf. ↑	FID ↓	Extreme Latents ($\lvert z \rvert > 2.2$)	Qualitative Synthesis Be
$\alpha = 1.0$ (Standard Cond.)	94.0%	93.1%	**28.08**	**4.52%**	Natural stroke variation; c class boundaries
$\alpha = 3.0$ (Moderate CFG)	**100.0%**	**99.9%**	71.13	5.61%	**Optimal balance:** 100% ence & crisp digits
$\alpha = 7.5$ (Strong CFG)	99.5%	99.6%	94.26	7.46%	High contrast, canonical duced variance
$\alpha = 15.0$ (Extreme CFG)	99.5%	96.9%	142.31	10.55%	**Over-guided saturation:** & boundary artifacts

Key Insight: Moderate guidance ($\alpha \in [3.0, 7.5]$) acts as a semantic amplifier that eliminates class ambiguity ($94.0\% \rightarrow 100.0\%$ prompt accuracy), whereas excessive extrapolation ($\alpha = 15.0$) pushes $\tilde{\epsilon}_\theta$ outside the training latent support ($\lvert z \rvert > 2.2$ doubles to 10.55%).

9. Pixel-Space vs. Latent-Space Diffusion Comparison (Task 6)

Matched Backbone Efficiency Benchmark (Model A: $32 \times 32 \times 1$ Pixel U-Net vs. Model B: $8 \times 8 \times 4$ Latent U-Net)

- Both models share identical channel widths ($64 \rightarrow 128$), ResBlocks, 4-head spatial self-attention at the bottleneck, batch size (128), and AdamW optimizer across 6 epochs on the same NVIDIA Tesla T4 GPU: **Model B trains $7.61\times$ faster (6.75 s/epoch vs. 51.42 s/epoch) and uses $13.81\times$ less peak VRAM (83.4 MB vs. $1{,}151.7$ MB).**

Concise Evidence-Supported Comparison Across Six Explicit Criteria

Explicit Criterion	Pixel-Space Diffusion (DDPM / Model A)	Latent Diffusion Model (LDM / Model B)
1. Operating Space	Full pixel tensor $\mathbf{x}_t \in \mathbb{R}^{H \times W \times C}$ ($32 \times 32 \times 1$, $1{,}024$ spatial locations).	Compressed latent $\mathbf{z}_t \in \mathbb{R}^{(H/f) \times (W/f) \times c}$ ($8 \times 8 \times 4$, $f = 4$, 64 locations; $16\times$ smaller grid).
2. Attention & Memory	$\mathcal{O}((H \cdot W)^2)$ self-attention; **1,151.7 MB** peak training VRAM on T4 GPU.	$\frac{1}{f^4}\mathcal{O}((H \cdot W)^2)$ ($256\times$ lower attention cost); **83.4 MB** peak VRAM ($13.81\times$ **reduction**).
3. Training Speed	Slow: **51.42 s/epoch** ($1{,}167$ img/s; 308.5 s total for 6 epochs).	Fast: **6.75 s/epoch** ($8{,}882$ img/s; 40.5 s total for 6 epochs, $7.61\times$ **faster**).
4. Sampling Throughput	Slow: T full-resolution U-Net passes (3.93 s per 64 MNIST images at $T = 300$).	Fast: T lightweight passes on compact $\mathbf{z}_t$ + 1 fast decoder pass $\mathcal{D}(\tilde{\mathbf{z}}_0)$.
5. Sample Quality	High at low resolution (MNIST FID **16.85**; CIFAR-10 FID **3.17** [1]).	High at high resolution (CelebA-HQ FID **5.11** [2]; Stage-1 VAE reconstruction FID **0.18**).
6. Conditioning & Limit	Rigid spatial concatenation; memory-bound at megapixel resolutions (512^2+).	Flexible cross-attention ($\tau_\theta(y)$ achieves 99.5%–100% acc. at $\alpha \in [3.0, 7.5]$); bounded by frozen VAE $\mathcal{D}$.

10. Critical Analysis, Practical Limitations & Conclusion (Task 6)

Technical & Ethical Limitations (Task 6 & Rubric Criterion 5)

1. **Sequential Sampling Latency:** Markovian reverse chains require tens to hundreds of sequential U-Net evaluations. Deterministic ODE solvers (DDIM [8], DPM-Solver [10]) and progressive distillation [11] reduce sampling to 1–50 steps, but aggressive step truncation can trade off fine texture or mode coverage.
2. **Prompt Binding & CFG Oversaturation:** Cross-attention models occasionally fail at multi-object attribute binding or spatial counting, and excessive CFG scales ($\alpha = 15.0$) induce latent saturation and contrast clipping.
3. **Training Data Memorization & Copyright:** Carlini et al. [12] demonstrated that large diffusion models can memorize duplicated training images and regurgitate near-exact copies of copyrighted works under specific prompts.
4. **Societal Bias & Representational Disparity:** Unfiltered web-scale datasets encode demographic and cultural stereotypes that conditional diffusion models amplify unless mitigated via dataset curation and debiasing.

Scope & Conditions of Applicability

- **Avoiding Overstatement:** Our controlled benchmarks use 28×28 and 32×32 MNIST digits on a single Tesla T4 GPU to isolate architectural scaling laws.
- At megapixel resolutions (512^2–1024^2), pixel-space $\mathcal{O}((H \cdot W)^2)$ self-attention becomes prohibitive, making LDM compression indispensable—provided the Stage-1 VAE is pre-trained on rich natural textures so fine details are not lost in $\mathcal{E}(\mathbf{x})$.

General Conclusion

In conclusion, reformulating generation as variance-preserving Gaussian corruption followed by iterative noise prediction enables diffusion models to achieve both sharp sample fidelity and full mode coverage. By moving the diffusion trajectory into a perceptual VAE latent manifold equipped with cross-attention, Latent Diffusion Models overcome the quadratic spatial bottleneck of pixel-space diffusion and make controllable synthesis practical.

References, Verification Statement & Discussion (Q&A)

Key References

[1] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," *NeurIPS*, vol. 33, pp. 6840–6851, 2020. (**Required Paper**)

[2] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-Resolution Image Synthesis With Latent Diffusion Models," *CVPR*, pp. 10684–10695, 2022. (**Recommended Paper**)

[3] J. Sohl-Dickstein et al., "Deep Unsupervised Learning using Nonequilibrium Thermodynamics," *ICML*, pp. 2256–2265, 2015.

[4] D. P. Kingma and M. Welling, "Auto-Encoding Variational Bayes," *ICLR*, 2014.

[5] I. Goodfellow et al., "Generative Adversarial Nets," *NeurIPS*, vol. 27, pp. 2672–2680, 2014.

[6] A. van den Oord, N. Kalchbrenner, and K. Kavukcuoglu, "Pixel Recurrent Neural Networks," *ICML*, pp. 1747–1756, 2016.

[7] Y. Song and S. Ermon, "Generative Modeling by Estimating Gradients of the Data Distribution," *NeurIPS*, vol. 32, 2019.

[8] J. Song, C. Meng, and S. Ermon, "Denoising Diffusion Implicit Models," *ICLR*, 2021.

[9] Z. Xiao, K. Kreis, and A. Vahdat, "Tackling the Generative Learning Trilemma with Denoising Diffusion GANs," *ICLR*, 2022.

[10] C. Lu et al., "DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps," *NeurIPS*, 2022.

[11] T. Salimans and J. Ho, "Progressive Distillation for Fast Sampling of Diffusion Models," *ICLR*, 2022.

[12] N. Carlini et al., "Extracting Training Data from Diffusion Models," *USENIX Security*, pp. 5253–5270, 2023.

Verification & Reproducibility

- **Dataset & Hardware:** Torchvision MNIST ($60{,}000$ train / $10{,}000$ test); all models trained and timed on an NVIDIA Tesla T4 GPU (CUDA 13.0, PyTorch).
- **Independent Verification:** All mathematical derivations, 1D numerical calculations, and empirical metrics were independently verified via reproducible scripts in `https://github.com/ChiVan222/AI_Seminar_Diffusion` (AI coding assistants used for LaTeX formatting and plotting).

Thank You — Q&A (5 mins)

Vong Chi Van (24125049)
Le Dat Phuc An (24125052)